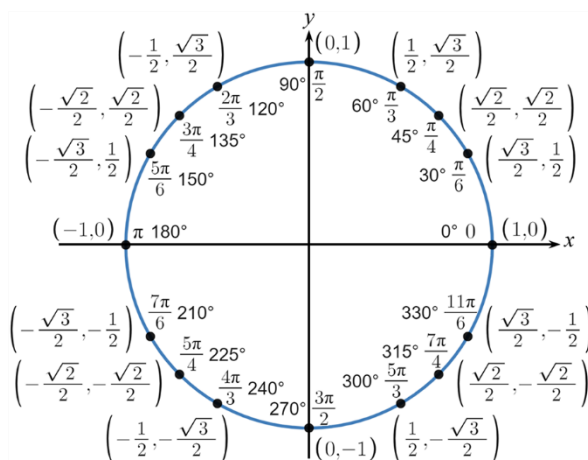


Graphing Sine and Cosine

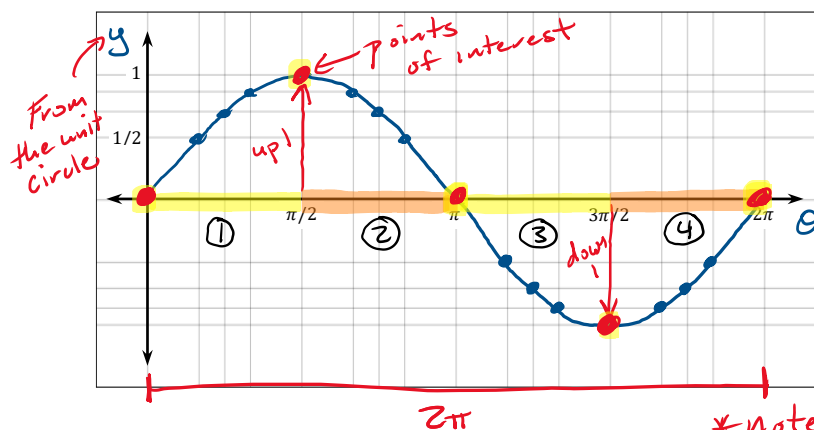
Circular versus Rectangular Graphs

Thus far, we've considered a strictly circular graph for the sine and cosine functions, namely the unit circle. In this lesson, we'll transition to the more familiar rectangular perspective typically associated with graphing. To achieve this, we'll consider our angle θ as the input and graph this independent variable along the horizontal axis. The corresponding output will be graphed along the vertical axis as usual.



Graph of the Sine Function

Graph the sine function by plotting angles from the unit circle along the horizontal axis and the corresponding output (y-values from the unit circle) along the vertical axis.

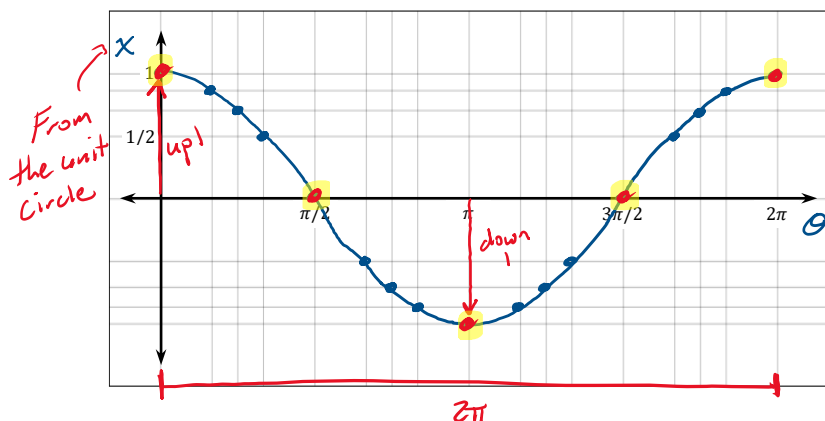


Domain: $(-\infty, \infty)$ any amount of rotation
 Range: $[-1, 1]$
 Amplitude: 1
 Period: 2π
 y-intercepts: $(0, 0)$
 x-intercepts: $(\pi k, 0)$, k any integer

*note each cycle has $\leq$ points of interest which partitions the period into 4 equal subintervals

Graph of the Cosine Function

Graph the cosine function by plotting angles from the unit circle along the horizontal axis and the corresponding output (x-values from the unit circle) along the vertical axis.



Domain: $(-\infty, \infty)$ any amount of rotation
 Range: $[-1, 1]$
 Amplitude: 1
 Period: 2π
 y-intercepts: $(0, 1)$
 x-intercepts: $(\frac{\pi}{2} + \pi k, 0)$, k any integer

Amplitude

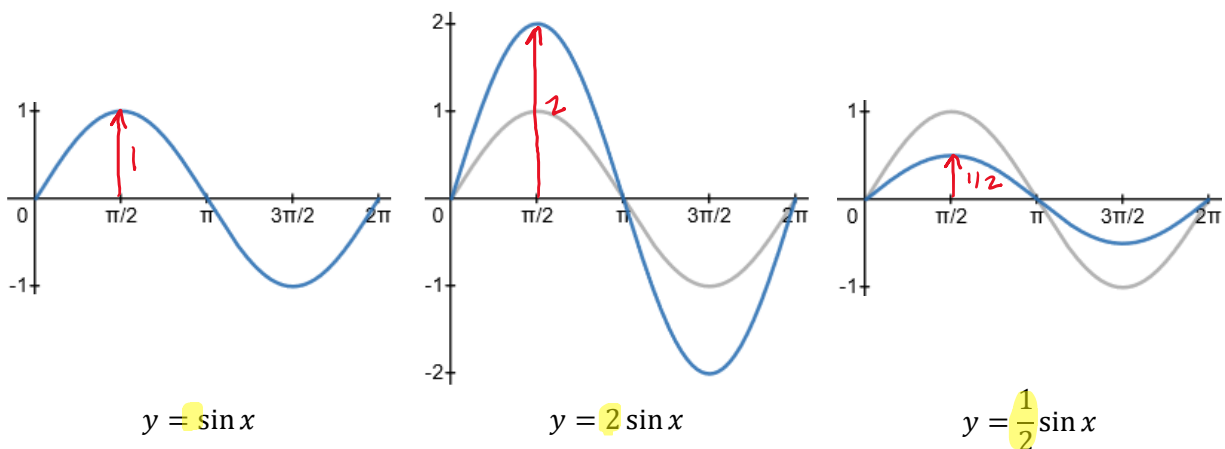
When graphing trigonometric functions, amplitude is a term that applies only to sine and cosine and refers to the rise and fall that we see in the graph.

DEFINITION: *Amplitude*

For the sine and cosine functions, the **amplitude** is the vertical distance between the maximum or minimum value along the curve (peaks and valleys) and the sinusoidal (or horizontal) axis.

Note when $y = a \sin x$ or $y = a \cos x$ the amplitude is equal to $|a|$. That is, amplitude is always a positive value. If $a < 0$, then the graph is reflected across the sinusoidal axis.

Here we see a direct relationship between amplitude and the multiplier in front of the function.



EXAMPLE 1

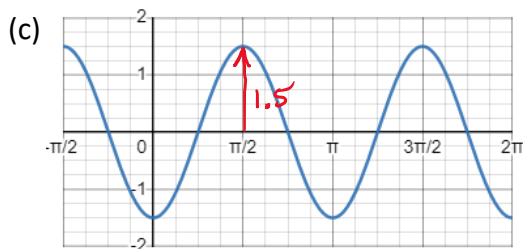
Determine the amplitude for each function below.

(a) $y = 3 \sin(2x) - 1$

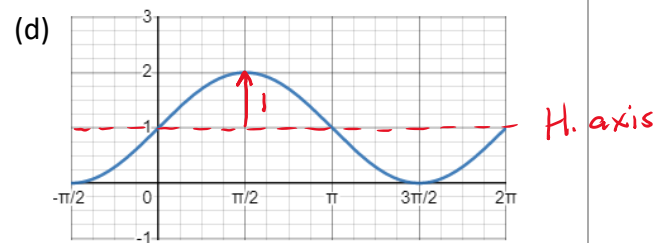
$\text{amp} = |3| = 3$

(b) $y = \frac{1}{2} - 2 \cos(\pi + 3x)$

$\text{amp} = |-2| = 2$



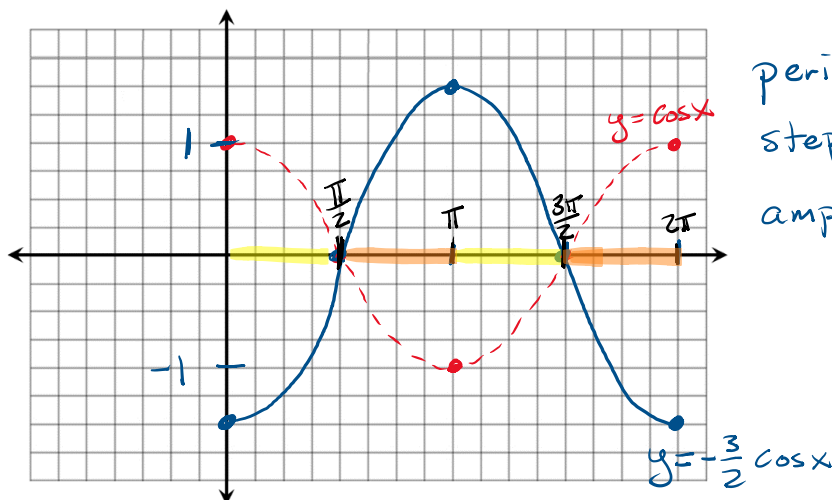
$\text{amp} = 1.5$



$\text{amp} = 1$

EXAMPLE 2

Graph $y = -\frac{3}{2}\cos x$ and $y = \cos x$ on the same coordinate axes.



$\text{period} = 2\pi$
 $\text{step} = \frac{\text{period}}{4} = \frac{2\pi}{4} = \frac{\pi}{2}$
 $\text{amp} = \left| -\frac{3}{2} \right| = \frac{3}{2}$
 reflect across H. axis

Period

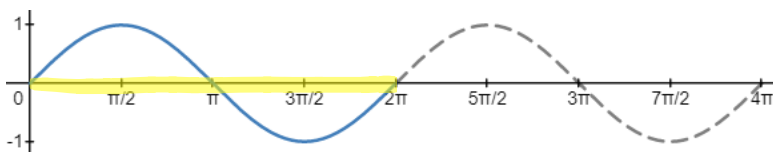
Functions like sine and cosine are cyclic with a definite repeating pattern. These are called periodic functions, and period is a term we'll use to describe this behavior.

DEFINITION: Period and Periodic Functions

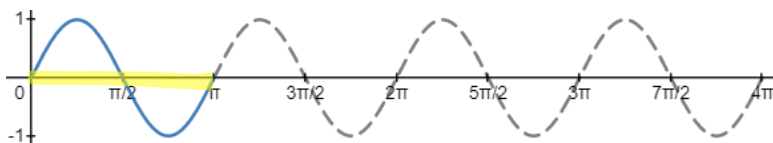
A **periodic function** is any function with a repeating pattern at regular intervals. The horizontal length that it takes to complete one full cycle is called the **period**.

Note when $y = \sin(bx)$ or $y = \cos(bx)$, with $b > 0$, the period is equal to $2\pi/b$.

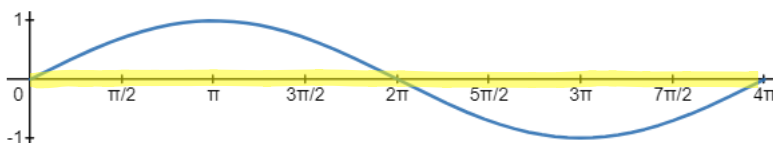
See how the examples below agree with the rule described in the note above.



$y = \sin(x)$ Period = 2π
 $\frac{2\pi}{1}$



$y = \sin(2x)$ Period = π
 $\frac{2\pi}{2}$



$y = \sin\left(\frac{1}{2}x\right)$ Period = 4π
 $\frac{2\pi}{1/2}$

EXAMPLE 3

Determine the period for each function below.

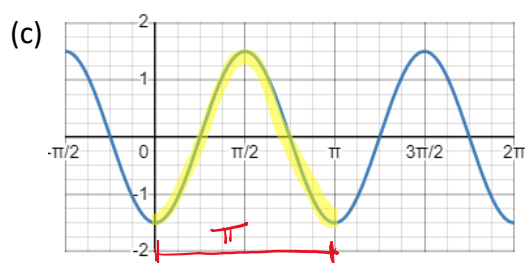
(a) $y = 3 \sin(2x) - 1$

$$\text{period} = \frac{2\pi}{2}$$

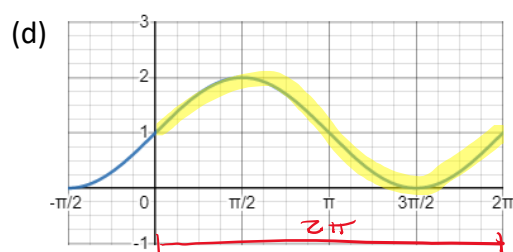
$$= \pi$$

(b) $y = \frac{1}{2} - 2 \cos(\pi + 3x)$

$$\text{period} = \frac{2\pi}{3}$$



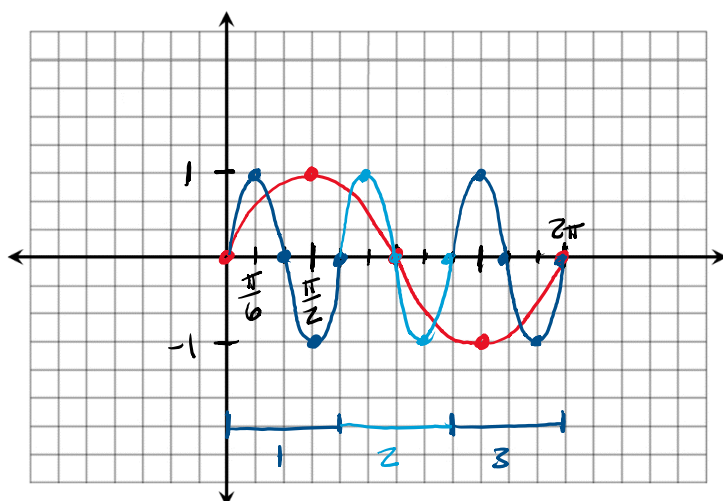
$$\text{period} = \pi$$



$$\text{period} = 2\pi$$

EXAMPLE 4

Graph $y = \sin(3x)$ and $y = \sin(x)$ on the same coordinate axes.



$$y = \sin(x)$$

$$\text{period} = 2\pi$$

$$\text{step} = \frac{2\pi}{4} = \frac{\pi}{2} \rightarrow \frac{3\pi}{6}$$

(common denominator)

$$y = \sin(3x)$$

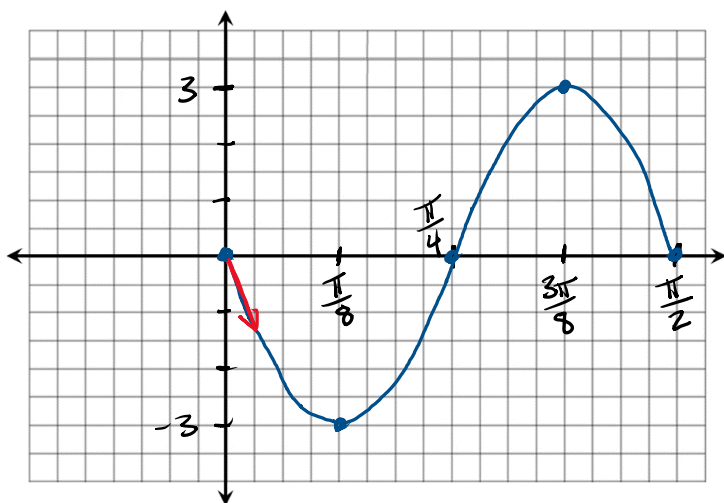
$$\text{period} = \frac{2\pi}{3}$$

$$\text{step} = \frac{2\pi/3}{4} = \frac{2\pi}{12} = \frac{\pi}{6}$$

EXAMPLE 5

Identify the amplitude and period, then graph each function below.

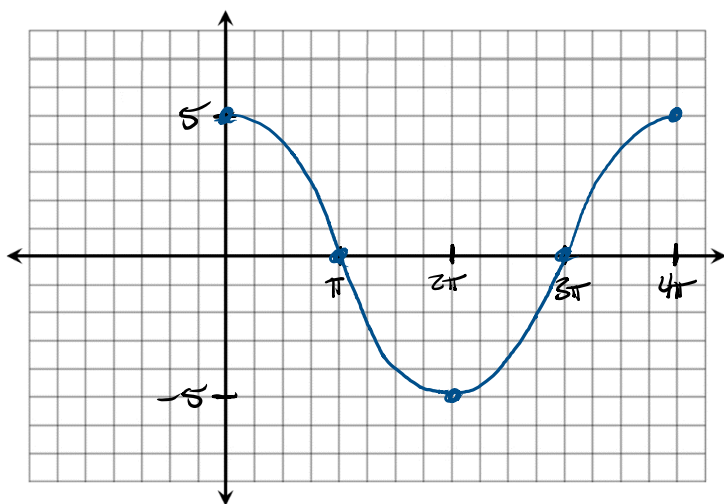
(a) $y = -3 \sin(4x)$



$$y = -3 \sin(4x)$$
$$\text{period} = \frac{2\pi}{4} = \frac{\pi}{2}$$
$$\text{step} = \frac{\pi/2}{4} = \frac{\pi}{8}$$
$$\text{amp} = |-3| = 3$$

↑ reflect across
H. axis

(b) $f(\theta) = 5 \cos\left(\frac{\theta}{2}\right)$



$$f(\theta) = 5 \cos\left(\frac{\theta}{2}\right)$$
$$\text{period} = \frac{2\pi}{1/2} = 4\pi$$
$$\text{step} = \frac{4\pi}{4} = \pi$$
$$\text{amp} = |5| = 5$$

